PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Programming Fundamentals Lab 06	**Instructor: Mr. Muhammad Hamza**
Program: AI Fall 2025	**Date: 16th November, 2025**
Marks: 10	**Lab Engineer: Qandil Fatima**
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Instructions:

- Submit the file with your names following your registration numbers like **AI001_Name.**
- **Copy Paste work will be given no marks.**

Lab Task

1. Program to print all natural numbers from 1 to n using while loop.

Untitled1.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    int i=1;
cout<<" Enter n:";
cin>> n;
while (i <= n)
{
    cout<<i<<" "<<endl;
i++;
}
return 0;
}
```

Explanation:

- Start i from 1 and keep printing i until it exceeds n.
- The loop checks if i <= n; if true, it prints i then increase i by one.
- Stops after printing up to n.

Output:

C:\Users\DELL PRECision 7550

```
 Enter n: 3
1
2
3

--------------------------------
Process exited after 2.306 seconds with return value 0
Press any key to continue . . .
```

2. Program to print natural numbers in reverse from n to 1 using while loop.

Untitled1.cpp [*] Untitled2

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cout << "Enter n: ";
    cin >> n;
    while (n >= 1)
    {
        cout << n << " "<<endl;
        n--;
    }
    return 0;
}
```

Explanation:

- Starts with n and prints it, then decreases n.
- The loop prints numbers in reverse, ending when n is less than 1.

Output:

C:\Users\DELL PRECision 7550

```
Enter n: 6
6
5
4
3
2
1

--------------------------------
Process exited after 3.728 seconds with return value 0
Press any key to continue . . .
```

3. Program to print all even numbers between i to n using while loop.of Natural Numbers using loop.

[*] Untitled1 [*] Untitled2

```
#include <iostream>
using namespace std;

int main()
{
    int i, n;
    cout << "Enter starting value i: ";
    cin >> i;
    cout << "Enter ending value n: ";
    cin >> n;
    while (i <= n)
    {
        if (i % 2 == 0)
            cout << i << " ";
        i++;
    }
    return 0;
}
```

Explanation:

- Starts from user-defined i and prints only even numbers (i % 2 == 0).
- Increments i each time, stopping at n.

Output:

C:\Users\DELL PRECision 7550

```
Enter starting value i: 8
Enter ending value n: 2

--------------------------------
Process exited after 23.8 seconds with return value 0
Press any key to continue . . .
```

4. Program to Generate Fibonacci Sequence up to a Certain Number. The Fibonacci sequence is a series of numbers where a number is found by adding up the two numbers before it. Starting with 0 and 1, the sequence goes 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, and so forth.

[*] Untitled1 Untitled2.cpp [*] Untitled3

```cpp
#include <iostream>
using namespace std;

int main()
 {
    int n, t1 = 0, t2 = 1, nextTerm = 0;
    cout << "Enter n: ";
    cin >> n;
    cout << t1 << " " << t2 << " ";
    nextTerm = t1 + t2;
    while (nextTerm <= n) {
        cout << nextTerm << " ";
        t1 = t2;
        t2 = nextTerm;
        nextTerm = t1 + t2;
    }
    return 0;
}
```

Explanation:

- Starts with first two Fibonacci numbers: 0, 1.
- Calculates the next number by adding the previous two.
- Prints terms until the next term exceeds n.

Output:

C:\Users\DELL PRECision 7550

```
Enter n: 34
0 1 1 2 3 5 8 13 21 34
--------------------------------
Process exited after 9.895 seconds with return value 0
Press any key to continue . . .
```

5. Take i and n two inputs from user and print n^i of the number.

[*] Untitled1 Untitled2.cpp Untitled3.cpp Untitled4.cpp Untitled5.cpp Untitled6.cpp Untitled7.cpp

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
 {
    int n, i;
    cout << "Enter n: ";
    cin >> n;
    cout << "Enter power i: ";
    cin >> i;
    cout << n << "^" << i << " = " << pow(n, i);
    return 0;
}
```

Explanation:

- Uses the built-in pow function to calculate n raised to the power i.
- Prints the result.

Output:

```
C:\Users\DELL PRECision 7550

Enter n: 4
Enter power i: 6
4^6 = 4096
--------------------------------
Process exited after 19.43 seconds with return value 0
Press any key to continue . . .
```

6. Write a C++ program using a while loop that computes the sum of first n positive integers: sum = 1 + 2 + 3 + … + n.

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | Untitled4.cpp | [*] Untitled5

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n, sum = 0, i = 1;
    cout << "Enter n: ";
    cin >> n;
    while (i <= n)
    {
        sum += i;
        i++;
    }
    cout << "Sum of first " << n << " positive integers is " << sum;
    return 0;
}
```

Explanation:

- Adds each number from 1 to n to sum.
- Prints the total sum after the loop completes.

Output:

```
C:\Users\DELL PRECision 7550

Enter n: 6
Sum of first 6 positive integers is 21
--------------------------------
Process exited after 3.518 seconds with return value 0
Press any key to continue . . .
```

7. Write a C program using do-while loop that asks the user to enter two numbers. The numbers should be added and the sum displayed. The user should be asked if he or she wishes to perform the operation again. If so, the loop should repeat; otherwise it should terminate.

Here is a **C program** that uses a do-while loop to repeatedly ask the user for two numbers, adds them, displays the sum, and asks if the user wants to repeat:

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | Untitled4.cpp | Untitled5.cpp | Untitled6.cpp | Untitled7.cpp | [*] Untitled8

```c
#include <stdio.h>

int main() {
    int num1, num2, sum;
    char choice;

    do {
        // Ask user to enter two numbers
        printf("Enter first number: ");
        scanf("%d", &num1);
        printf("Enter second number: ");
        scanf("%d", &num2);

        // Add the numbers and display the sum
        sum = num1 + num2;
        printf("Sum = %d\n", sum);

        // Ask user if they want to continue
        printf("Do you want to perform the operation again? (y/n): ");
        // Note: space before %c consumes any leftover newline character
        scanf(" %c", &choice);

    } while (choice == 'y' || choice == 'Y');

    return 0;
}
```

Explanation:

- Uses do-while loop so that the operation is performed at least once.
- Inside the loop:
  - Prompts the user to enter two integers.
  - Adds the two numbers and prints the result.
  - Asks the user if they want to repeat.
- If the user enters 'y' or 'Y', the loop starts over; otherwise, it ends.
- The " %c" in scanf ensures it correctly captures the user's choice, even if you press Enter after a number.

Output:

```
C:\Users\DELL PRECision 7550

Enter first number: 6
Enter second number: 5
Sum = 11
Do you want to perform the operation again? (y/n): y
Enter first number: 2
Enter second number: 3
Sum = 5
Do you want to perform the operation again? (y/n): n

--------------------------------
Process exited after 12.33 seconds with return value 0
Press any key to continue . . .
```

8. Write a C program to Convert the following while loop to a do-while loop:

```
int x = 1;
 while (x > 0)
 {
cout << "enter a number: ";
 cin >> x;
 }
```

[*] Untitled1 | Untitled2.cpp | Untitled3.cpp | Untitled4.cpp | Untitled5.cpp | Untitled6.cpp

```c
#include <stdio.h>

int main()
 {
    int x = 1;
    do
     {
        printf("Enter a number: ");
        scanf("%d", &x);
    }
     while (x > 0);
    return 0;
}
```

Explanation:

- The loop asks the user to enter a number and stores it in x.
- The body runs **at least once** because the condition is checked after the first execution.
- If the entered number is greater than 0, it repeats; otherwise, it exits.
- This is a classic use of a do-while loop for user input validation/repetition.

Difference:

- The while loop checks the condition first, so it may *never* run if x > 0 is false initially.
- The do-while loop **always executes once** before checking the condition, making it ideal for prompting the user at least one time.

Output:

C:\Users\DELL PRECision 7550

```
Enter a number: 2
Enter a number: 3
Enter a number: 4
Enter a number: 5
Enter a number: 6
Enter a number: 7
Enter a number: 8
Enter a number: 9
Enter a number: 0

--------------------------------
Process exited after 8.486 seconds with return value 0
Press any key to continue . . .
```